Homework 8

October 30, 2026

Problem 1

Let

$$A = \begin{pmatrix} -7 & 16 & 8 \\ -4 & 9 & 4 \\ -4 & 7 & 5 \end{pmatrix}.$$

Find out the minimal polynomial of A.

Problem 2

Let

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -4 & -4 \end{pmatrix}.$$

Find out the minimal polynomial of A.

Problem 3

In Problem 3 of Homework 7, A was a matrix whose elementary divisors are

$$(\lambda+3), \quad (\lambda+3), \quad (\lambda+3)^2, \quad \lambda+2+5j, \quad \lambda+2-5j, \quad (\lambda+2+5j)^2, \quad (\lambda+2-5j)^2.$$

Based on that result, find out the minimal polynomial of A.

Problem 4

In Problem 4 of Homework 7, A was a matrix whose elementary divisors are

$$(\lambda+1), \quad (\lambda+1)^2, \quad (\lambda+1+j), \quad (\lambda+1-j), \quad (\lambda+1)^2, \quad (\lambda+1+j)^2, \quad (\lambda+1-j)^2.$$

Based on that result, find out the minimal polynomial of A.

Problem 5

Let $\mathscr{A} \in L(C_n[\lambda])$ be defined as

$$\mathscr{A}[f(\lambda)] = f(\lambda + a), \qquad \forall\, f(\lambda) \in C_n[\lambda],$$

for a nonzero constant $a \in C$. Find out

(1) all the generalized eigenvectors of $\mathscr{A}$;

(2) the transition matrix X that reduces A to the Jordan canonical form $J = X^{-1}AX$;

(3) the generating elements $\beta_i(s)$, $i = 1, 2, \ldots, l$, such that

$$C_n[s] = \mathscr{S}_{\mathscr{A}}[\beta_1(s)] \dot{+} \mathscr{S}_{\mathscr{A}}[\beta_2(s)] \dot{+} \cdots \dot{+} \mathscr{S}_{\mathscr{A}}[\beta_l(s)].$$